

Far Western University
Bachelor of Engineering (Civil)
Course of Study 2071

Course Title: Soil and Rock Mechanics

CourseCode.: CE361

3Year/Semester: Third/Sixth

Level: Bachelor of Engineering (Civil) Tutorial/week: 1

Credit: 3

Number of lecture/week:

Total hours: 45

1. Course Introduction:

This course is aimed at teaching the students the concepts of soil and rock engineering, including the science and technology of soils and rocks and their application to problems in civil engineering.

2. Course Objectives:

At the end of this course the student should be able to:

- Emphasizes the fundamentals and relevant principles of soil mechanics,
- Give an overall picture of the behaviour of soils
- Describe the nature of some of the soil problems encountered in civil engineering
- To understand the basic concept and tools of rock mechanics in the field of geotechnical engineering.

3. Specific Objectives and Contents:

Specific Objectives	Contents
• Understand meaning and scope • Understand different soil formation process	UNIT 1. Introduction (1hour) 1.1 Preview of Geotechnical problems in civil Engineering and infrastructure Development. 1.2 Historical development of soil mechanics. 1.3 Soil formation and soil type.
• Understand phase diagram for different soil conditions • Understand different index properties and laboratory determination	UNIT 2. Phase Relations and Index properties of soils(4 hours) 2.1 Phase diagram 2.2 Simple definitions and their relationship 2.3 Determinations of various index properties

Specific Objectives	Contents
- Understand different soil classification system - Understand the applicability of different classification system	UNIT 3. Soil Classification (2 hours) 3.1 Introduction 3.2 Soil classification-Textural, ISSCS, MIT, BSCS, USCS and AASHTO soil classification system 3.3 Application of soil classification system
- Understand the compaction process - Understand the different compaction process in the lab and field	UNIT 4. Soil Compaction (2hours) 5.1 Introduction 5.2 Laboratory tests 5.3 Factors affecting compaction 5.4 Compaction specification and field control.
- Understand the concept of Permeability and effective stress - Understand the different laboratory and field methods of permeability - Understand the concept of seepage and its field applicability	UNIT 5.Principle of Effective Stress, Permeability and seepage analysis (6hours) 6.1 Introduction 6.2 Physical meaning of effective stresses 6.3 Capillarity in soils 6.4Permeability of soils 6.5 Determinations of coefficient of permeability: Laboratory and field methods. 6.6 Seepage forces and quick sand conditions 6.7 Seepage analysis in isotropic and anisotropic soil conditions
- Understand the concept of stress distribution - To familiarise the different stress distribution systems in different conditions	UNIT 6.Stress Distribution of Soils(3 hours) 6.1 Introduction 6.2 Boussinesq equations and Westergaards equation 6.3 Vertical Stress Distribution Diagrams 6.4 New marks influence chart 6.5 Approximate stress distribution methods for Loaded Areas
- Understand the concept of consolidation behavior of soil - To familiarise the compressibility characteristics of different soil - Understand the concept of calculating different settlement analysis	UNIT 7.Compressibility of soil and consolidation(6 hours) 7.1Fundamentals of Consolidation 7.2One –Dimensional Laboratory consolidation Test 7.3 Void Ratio – Pressure plots 7.4Normally consolidated and over consolidated clay 7.5 Calculation of Settlement from One – Dimensional Primary Consolidation 7.6 Secondary Consolidation Settlement 7.7 Coefficient of Consolidation 7.8Calculation of Consolidation Settlement under a Foundation 7.9Method of Accelerating Consolidation Settlement

Specific Objectives	Contents
- Understand the basic soil parameter - Familiarise with different lab and field test to calculate the shear strength parameter of soil	UNIT8. Shear Strength of Soil (5 hours) 8.1 Mohr-Coulomb failure criterion 8.2 Laboratory Tests For Determination of shear strength Parameters. 8.3 Direct Shear Test 8.4 Triaxial Shear Test-General 8.5 Consolidated drained Triaxial Test 8.6 Consolidated undrained Triaxial Test 8.7 Unconsolidated undrained Triaxial Test 8.8 Unconfined compression Test on Saturated clay. 8.9 Stress Path 8.10 Vane Shear Test 8.11 Shear Strength of Sands.
To familiarise with the different soil stability analysis technique	UNIT9. Slope Stability Analysis (4 hours) 9.1 Introduction 9.2 Infinite slopes and Translation slides 9.3 Definition of factor of safety 9.4 Finite slopes- Forms of Slip surface 9.5 $\phi = 0$ Analysis (Total stress Analysis) 9.6 $c - \phi$ Analysis – Method of Slices. 9.7 Friction Circle Method 9.8 Taylor's Stability Number 9.9 Bishop's method of Stability Analysis 9.10 Use of Stability Coefficients
Understand the basic properties of rocks	UNIT10. Classification and Index properties of Rocks (4 hours) 10.1 Geological classification 10.2 Index properties of rock system 10.3 Porosity, density, Permeability 10.4 Slaking and durability

Specific Objectives	Contents
Understand about the rock strength and different failure criteria	UNIT11. Rock Strength and Failure Criteria (2 hours) 11.1 Modes of failure of rock 11.2 Common lab strength test 11.3 σ - ϵ behavior in compression 11.4 Mohr-Coulomb failure criterion
Understand the different stress condition and deformation characteristics	UNIT12. Stresses and Deformability of Rocks(2 hours) 12.1 Initial stress, horizontal stress and vertical stress 12.2 Techniques for measurements of in-situ stresses 12.3 Measurements of deformability(lab compression test, plate bearing test, borehole and gallery test, flat jack test, dynamic measurements)
Understand the application of Rock Mechanics to Rock Slope Engineering	UNIT13. Application of Rock Mechanics to Rock Slope Engineering(4 hours) 13.1 Modes of failure of slope in hard rock 13.2 Kinematic analysis of slopes 13.3 Analysis of plane sliding of the stereographic projection 13.3 Analysis of wedge sliding using stereographic projection 13.4 Analysis of slides composed of two blocks

Laboratories

- Sieve analysis of coarse and fine grained soils.
- Determination of Atterberg limit of soils
- Determination of In-situ density by Sand replacement method and Core Cutter Method.
- Determination of OMC and maximum dry density
- Unconfined compression test
- Direct shear Test
- Constant head permeability Test
- UU Triaxial Test

References:

- "Soil mechanics in Engineering Practice" Terzaghi K and Peck. R. B. John Wiley New York, 1967
- "Principles of Geotechnical Engineering" Braja M. Das Fifth edition. Thomson/Brooks
- "Physical and Geological properties of soils", Joseph E Bowles, McGraw Hill Edition 1984.
- "Basic and Applied soil mechanics "GopalRanjan and ASR Rao, second edition New Age International publishers, 2000
- "Soil mechanics and foundation Engineering" K. R. Arora Standard Publisher 1997
- "A Text Book of Soil Mechanics and Foundation Engineering in SI units "V.N.S. UBS Publishers Distributors Ltd. Fourth edition 1993
- "Engineering in Rocks for Slopes, Foundations and Tunnels" T. Ramamurthy, Eastern Economy Edition, Prentice Hall of India Pvt. Ltd., 2007
- "Rock Mass Classification, A Practical Approach in Civil Engineering", Singh, B, Elsevier Science Ltd, First edition 1999

Far Western University
Bachelor of Engineering (Civil)
Course of Study 2073

Course Title Irrigation Engineering

Credit :3

Course No : CE362

Number of Hours per week(2+1)

Nature of the Course : Theory + Tutorial

Total Hours 45

Practical : 1.5 / 2 Hour each week

Level : Bachelor of Engineering (Civil)

Year : Third

Semester : Second

1. Course Introduction:

This course is advance course for Bachelor students. Before teaching this course, students should know the basic knowledge in fluid mechanics and hydraulics engineering. This course aimed to deliver the knowledge to the Civil Engineering Student of Third Year Second Part at Bachelor Level about the application of fluid mechanics, hydraulics engineering and hydrology inducing its own fundamental understanding. This course aims to deliver the knowledge to the students for demand analysis, planning, design, operation and maintenance of irrigation structures and its system components.

2. Course Objectives:

At the end of this course the students should be able -

- To understand the fundamental terms used in Irrigation Engineering.
- To know the behavior of Irrigation Engineering.
- To know the problems arise in Irrigation Engineering.
- To know the Overall design in irrigation Engineering (Design of individual components in irrigation engineering).

3. Specific Objectives and Contents:

Specific Objectives	Contents
➤ To know the concepts of irrigationengineering ➤ To know thenational cropping pattern ➤ To know the irrigation methods (current practice and possible extension in future)	Unit 1. Introduction (3 Hour) 1.1 History of Irrigation Engineering 1.2 Scope and advantages and disadvantages of irrigation 1.3 Cropping pattern in Nepal and cropping intensity 1.4 Command areas and Irrigation intensity 1.5 Field irrigation methods and their uses 1.6 Planning of irrigation projects
➤ To know the irrigation functions ➤ To know the terms duty, delta and cropping period and their relation and understanding ➤ To know the methods of irrigation water requirement ➤ To know the canal discharge necessary for irrigation	Unit 2 .Crop Water Requirements(5 Hours) 2.1 Functions of irrigation water 2.2 Duty, Delta and Crop periods 2.3 Factor affecting duty 2.4 Penman's Methods of crop water estimation 2.5 Soil-moisture-irrigation relationship 2.6 Consumptive use of Water (Evapo-Transpiration) and effective rainfall 2.7 Irrigation efficiencies and soil fertility 2.8 Discharge requirement in irrigation canals
➤ To know the irrigation by canals ➤ To know the types of canals use in irrigation ➤ To know the necessity of different types of canals	Unit 3 .Canal Irrigation System (3 Hours) 3.1 Principle of Canal irrigation 3.2 Classification of canals and their components 3.3 Alignment of different canals 3.4 Alluvial and non-alluvial canals 3.5 Necessity of main and branch canals
➤ To know the canal capacity use for irrigation	Unit 4 .Design of Canals(6 Hours) 4.1 Design of Canal Capacity

Specific Objectives	Contents
➤ To know the canal design principles and stability ➤ To know the canal design theory ➤ To know the economical canals use for irrigation	4.2 Sediment transport in canals 4.3 Tractive force approach of canal design 4.4 Design of stable canal 4.5 Design of Alluvial canals (Kennedy's and Lacey's Theory) 4.6 Design of lined canals with economical analysis
➤ To fix the position of irrigation headwork ➤ To know the headwork and its scope ➤ To understand the strength of headwork ➤ To be able to design the weir, and silt related structures	Unit 5 .DiversionHeadworks(7 Hours) 5.1 Necessity of Headwork and its location in irrigation 5.2 Weir/ Barrage with components with details 5.3 Causes of failure of weirs and their remedies 5.4 Seepage theory (Bligh's, Lane's and Khosla's theory) 5.5 Design of sloping glacis weir (crest, length and thickness of bed) 5.6 Design of Head Regulator, divide wall and fish ladder 5.7 Design of under-sluice and silt excluder 5.8 Design of silt ejector
➤ To understand the canal and off taking canals ➤ To know functions of regulating structures and its design ➤ To know the outlet and escape and its design principles ➤ To know the drops and its importance	Unit 6 .Water Regulating Structures(6Hours) 6.1 Difference between main canal and off-taking canals 6.2 Alignment of off-taking canals 6.3 Functions of regulating structures: Head regulator, Cross regulator, Outlet, Drop and Escapes 6.4 Design of Regulating structures and escapes (crest, length and thickness of floor) 6.5 Types of outlet and design of pipe outlet (free and submerged) 6.6 Types of drop, Design of vertical drop (crest, length and thickness of floor)
➤ To understand the cross drainage structures and its importance ➤ To be able to design the cross drainage structures	Unit 7 .Cross Drainage Structures(4 Hours) 7.1 Introduction of cross-drainage structures 7.2 Types of cross-drainage structures with drawing 7.3 Selection of cross-drainage structures 7.4 Design of Aqueducts and Syphon aqueducts (Detail drawing, drainage water way, barrel, canal water way, and transition, length and thickness of floor bed)
➤ To understand the basic of river engineering ➤ To know the nature of river flow path ➤ To know the river training works and its necessity in irrigation ➤ To be able to design the river training structures like Guide bunds, spurs etc.	Unit 8 .River Engineering and River Training (4 Hours) 8.1 Classification of River 8.2 Meandering and its causes 8.3 General features of meandering 8.4 River training and its necessity 8.5 Types of River training 8.6 Design of Guide bunds and launching apron 8.7 Design of Spurs (layout, length, spacing and cross section)
➤ To understand the drainage engineering ➤ To know water logging problems and its remedies ➤ To be able to design the surface and sub-surface drainage structures	Unit 9 .Draining Engineering (5Hours) 9.1 Introduction of Drainage engineering 9.2 Water logging in irrigation areas and importance of drainage 9.3 Causes, effects and preventative measures of Water logging 9.4 Surface design system and their design (layout of drainage, internal and external drainage, slope and cross section of drainage, re-structuring of existing drainage to improve for better drainage) 9.5 Sub-surface design system and their design (layout of drainage, slope and cross section of drainage)

Specific Objectives	Contents
➤ To understand the irrigation system ➤ To know the organization structure of irrigation ➤ To know the hill irrigation and its importance in Nepal	Unit 10 .Planning of Irrigation System (2 Hours) 10.1 Irrigation engineering systems planning 10.2 Irrigation Engineering system organization and management 10.3 Development of Small Irrigation projects 10.4 Introduction to Hill Irrigation

8 Practical / Tutorials :

The following Laboratory works or Tutorials will be performed during the course. If possible, practical work should be carried out in fully equipped laboratory, if not, concepts should be given through suitable tutorial works.

- 1 Exercise on Duty, Delta and Crop period
- 2 Exercise on crop water requirement
- 3 Exercise on soil-moisture-irrigation relation with irrigation interval
- 4 Exercise on economical canal design
- 5 Exercise on stable canal design and lined canal design
- 6 Exercise on design of Guide bunds and launching apron
- 7 Exercise on Khosla's theory for design of hydraulic structures
- 8 Exercise on design of sloping glacis weir bay
- 9 Exercise on Cross and head regulator design
- 10 Exercise on design of pipe outlet, vertical drop, siphon aqueduct
- 11 Exercise on surface and sub-surface drain design

9 Field Visit :

Two day field visit should be carried out to the students in suitable irrigation field.

After the field visit, students should submit the field visit report which includes:

- 1 Irrigation water requirement using CROPWAT software
- 2 Detail report of field visit study should be prepared individually and group presentation at the end

Far Western University
Bachelor of Engineering (Civil)
Course of Study 2071

Course Title: Design of Steel and Timber Structure
Course Code: CE363
Year/Semester: III/II
Level: BCE (Bachelor of Civil Engineering)

Number of lecture/ week: 4
Tutorial/ week: 2hrs
Total Lectures: 58hrs

1. Course Introduction

It is assumed that completion of this course student can design and supervise construct of general steel structures. They can design simple bolted/riveted and welded connections, flexure members, tension members, compression members and roof trusses. They also get the brief knowledge about timbers and timber structures.

2. Course Objectives

At the end of the course the students should be able to

- identify the steel and timber material behavior
- design and supervise of simple steel and timber structure; flexure members, tension members, compression members and roof trusses along with joints

3. Course Outline

Specific Objectives	Contents	Duration
Scope of the subject, Steel types, properties and use, Design philosophy and analysis	<u>Chapter 1: Introduction</u> • Types and properties of structural steels: standard quality, general structural steel, high tensile structural steel, grade designation, chemical composition, physical properties, mechanical properties • Use of steel as a structural member: Rolled steel sections, built-up sections • Stress-strain characteristics of structural steels: yield stress, ultimate stress, percentage elongation • Types of steel structures: Buildings, bridges, tower etc. • Method of Analysis and Design • Method of design, Design considerations, Simple design, semi-rigid design, fully rigid design & experimentally based design • Codes of practices and load combinations	4hrs
Basic design philosophy and application of working stress method	<u>Chapter 2: Working Stress Design Method</u> • Basic Assumptions in Working Stress Design • Service Load and Permissible Stresses • Design in Tension, Compression, Bending and Shear	2hrs

Specific Objectives	Contents	Duration
Basic design philosophy and application of working stress method	<u>Chapter 3: Limit State Design Method</u> • Safety and Serviceability Requirements of Structure • Different Limit States for Steel Design • Design Strength of Materials and Design Loads • Limit State of Strength • Limit State of Serviceability	3hrs
Basic principles and design of joints	<u>Chapter 4: Connections in Steel Structures</u> ○ Types of Connections ○ Welded Connections ▪ Welds and welding ▪ Design of simple welded connections ▪ Design of eccentric welded connections ○ Bolted Connections ▪ Bolts and bolting ▪ Design of simple bolted connections ▪ Design of eccentric bolted connections ▪ Introduction and design of Riveted Connection	10hrs
Basic principles and design of tension member	<u>Chapter 5: Tension Members</u> • Types of Tension Members • Sectional Area of Tension Membe • Design of Tension Members of Simple and Built-Up Section • Design of Lug Angle • Tension Splice	4hrs
Basic principles and design of compression member and its base, Basic design on joint of beam section	<u>Chapter 6: Compression Members</u> ○ Types and typical cross-section of compression member: column, stanchion, strut, standard section, built-up section ○ Buckling Behavior of Column - End conditions, effective length, slenderness ration and permissible stresses: Euler formula, Gordon-Rankine Formula ○ Strength of axially loaded compression member ○ Design of Column of Simple and Built-Up Section ○ Design of Lateral Bracing of Compression Member ○ Design of Eccentrically Loaded Column ○ Compression member subjected to bending. ○ Design of Column Bases ▪ Axially loaded column bases	10hrs

Specific Objectives	Contents	Duration
	▪ Eccentrically loaded column bases ○ Design of Column Splices	
Basic principles and design of flexure member and elements	<u>Chapter 7: Flexure Members</u> • Types of Beams • Design of Simple Beam -Effective span of beam, effective length of compression flange, allowable stresses in bending, bearing and shear. ○ Design of laterally supported beams& laterally unsupported beams: standard section, symmetrically built-up section and unsymmetrical built-up sections. ○ Design of Plate Girder ▪ Element of plate girders ▪ Preliminary design ▪ Design for bending, shear, deflection and lateral stability ▪ Curtailment of plate ▪ Design of beam end connection: framed connection and seated connection ▪ Design of web and flange splice	13hrs
Wind load and codal provision, Basic principles and design of roof truss and member	<u>Chapter 8: Design of Roof Trusses</u> • Types of Roof Truss and Components of Roof Truss • Basic wind speed, design wind speed, design wind pressure and wind load • External and internal wind pressure for slope roof, dead load and imposed load on roof, self weight of purlins, roof truss and wind bracing. • Loads on Roof Truss • Design of Roof Components - purlins: angular section, tubular section and I-sections • Design of angular and tubular truss members: strut, rafter and tie.	4hrs
Scope of the subject, Timber types, properties and use Design philosophy and analysis	<u>Chapter 9: Timber Structures and Design Methods</u> • Introduction to Timber Structures • Timber types and properties • Structural Timber and Factors Affecting the Strength of Timber • Design Methods and Basis for Design	2 hrs
Basic principles and design of different types of	<u>Chapter 10: Joints in Timber Structures</u> • 10.1. Types of Joints	2 hrs

Specific Objectives	Contents	Duration
joints	10.2. Design of Bolted Joints 10.3. Design of Nailed Joints	
Basic principles and design of compression and elements, Design of column base	Chapter 11: Design of Timber Compression Members 11.1. Types of Timber Columns 11.2. Design of Timber Column 11.3. Introduction to Column Bases	3 hrs
Basic principles and design of flexure, member and elements	<u>Chapter 12: Design of Timber Flexure Member</u> - Types of Beams - Design of Timber and Flitched Beam	3 hrs
Project work	Course Project: A Course Project on integrated design of a simple building structure	

Reference Materials:

1. "Limit State Design of Steel Structures" S.K. Duggal Tata McGraw-Hill Publishing Com.
2. "Design of Steel Structures" K.S. Sai Ram, PEARSON Education
3. "Design of Steel Structures" L.S. Negi, Tata McGraw-Hill Publishing Com.
4. "Design of Steel Structures" Ram Chandra, Standard Book House

Far Western University
Bachelor of Engineering (Civil)
Course of Study 2071

Course Title: Sanitary and Environmental Engineering Credit: 3
Course Code: CE 364 Number of lecture/week: 3
Year/Semester: Third / Sixth Tutorial/week:
Level: BCE (Bachelor of Civil Engineering) Lab/week:
Total Lectures: 45hrs

1. Course Introduction

The main aim of this course is to provide a basic knowledge to the students to understand the fundamentals of the sanitary and environmental engineering. The students will also study relationship of sanitary system and environment, fundamental principle of wastewater, quantification, quality assess and safe waste collection and disposal. The course covers the basic definition of terminology and design principle. The students will be able to analyze and design the different component of sanitary engineering and treatment works.

2. Course Objectives

At the end of this course the student should be able to:

- Understand sanitary system and functions of the various components, knowledge of wastewater on collection, conveyance, treatment, safe disposal, methods and design consideration.
- Have knowledge about rural sanitation, sludge management and introduction of solid waste management.

3. Course Outline

Specific Objectives	Contents	Duration in hours
· Introduction to fundamental of terms · Importance of environmental and sanitary engineering · Introduction of sanitation and sewerage	Unit 1. Introduction to sanitary and environmental engineering	2
	1.1. The impact of humans upon the environment and the impact of the environment upon humans	
	1.2. Definitions of common terms -Sewage/Wastewater, Domestic sewage, industrial sewage, Sanitary sewage, Storm water, Sullage, Sewer, Sewerage, Rubbish, Garbage, Refuse/Solid waste, environment , pollution	
	1.3. Importance of Wastewater and Solid Waste Management	
	1.4. Wastewater and Solid waste management methods Collection, Conveyance, Treatment and Disposal	

Specific Objectives	Contents	Duration in hours
system	1.5. Objectives of sewage disposal	
	1.6. Sanitation systems	
	1.6.1. Conservancy system with merits and demerits	
	1.6.2. Water carriage system with merits and demerits	
	1.7. Sewerage systems and types	
	1.7.1. Separate system	
	1.7.2. Combined system	
	1.7.3. Partially separate system	
	1.7.4. Comparison between separate and combined systems	
	1.8. Role of environmental Engineer	
- Knowledge to quantity estimation, role of rainfall in quantity and other influence factors - Numerical exercise	Unit 2. Quantity Estimation of Wastewater	3
	2.1. Dry Weather Flow (DWF) and Wet Weather Flow (WWF)	
	2.2. Sources of sanitary sewage	
	2.2.1. Private and public water supplies	
	2.2.2. Groundwater infiltration	
	2.2.3. Unauthorized connections	
	2.3. Factors affecting quantity of sanitary sewage	
	2.3.1. Population , rate of water supply, Groundwater infiltration ,Unauthorized connections	
	2.4. Determination of quantity of sanitary sewage, peak factor, peak flow	
	2.5. Different method of Determination of quantity of storm water	
	2.5.1. Rational method and its limitation, Overall runoff coefficient, British ministry of Health formula for intensity Of rainfall , Time of concentration	
	2.5.2. Numerical exercise on determination of quantity of wastewater for separate combined and partially separate systems	
- Knowledge to design of sewer, sewer material and construction procedure - Numerical exercise	Unit 3. Design and Construction of Sewers	4
	3.1. Criteria for sewer design	
	3.1.1. Necessary terms and terminology for design: Specific gravity of wastewater , Design period Minimum and Maximum velocities, Self-cleansing velocity ,Sewer size range, Sewer gradient, Hydraulic formulae for design Manning's, Chezy's and Hazen Williams formulae , Hydraulic elements of sewers for partial flow condition, Partial flow diagrams	
	3.2. Shapes of sewers : Circular and non-circular sections with merits and demerits	
	3.3. Common materials use for sewer : Requirements of sewer materials , Types of sewer materials salt glazed stoneware, cement concrete, cast iron	
	3.4. Design of sewers of separate and combined systems ,Numerical exercises on design of sewers	
	3.5. Construction of sewers	

Specific Objectives	Contents	Duration in hours
	3.5.1. Setting out, Alignment and gradient fixing, Excavation method of trench, Timbering of trench, Dewatering of trench, Laying and jointing, Testing of sewer Straightness, Obstruction, air tightness and water test, Steps of Backfilling of trench	
· Knowledge about appurtenances and material use and construction	Unit 4. Sewer Appurtenances	3
	4.1. Necessity of sewer appurtenances 4.2 Construction of sewer appurtenances	
	4.2. Main sewer appurtenances construction	
	4.2.1. Manhole, Drop manhole, Lamp hole, Street inlets, Catch basin, Flushing device, Sand, grease and oil traps, Inverted siphon, Sewer outlet	
· Knowledge of wastewater examination, sampling collection, characteristics of wastewater, and concept of BOD, COD etc.	Unit 5. Examination of waste water	5
	5.1. Characteristics of wastewater :	
	5.1.1. Physical Characteristics (colour, temperature, odour, turbidity)	
	5.1.2. Chemical Characteristics {pH, organic and inorganic solids, nitrogenous compounds}	
	5.1.3. Biological Characteristics {Bacteria}	
	5.2. Sampling of wastewater	
	5.2.1. Grab and composite samples, Preservation and storing,	
	5.3. Decomposition of wastewater-process,	
	5.3.1. Aerobic and anaerobic decomposition, Stale sewage, Biochemical Oxygen Demand (BOD), Definition of BOD and its significance, Derivation of BOD equation, Rate reaction, ultimate BOD and relation with temperature, Numerical on BOD, Chemical oxygen Demand (COD) Definition and significance, Examination of wastewater, Necessity of wastewater examination, Examination of volatile, fixed and total solids, settleable and non-settleable solids, BOD with and without dilution, COD, Numerical on BOD test	
· Knowledge about safe waste water disposal, its necessity and safe disposal methods. natural purification theory of river	Unit 6. Wastewater Disposal	5
	6.1. Necessity and objectives of wastewater disposal	
	6.2. Wastewater disposal methods Dilution and Land treatment	
	6.3. Wastewater disposal by Dilution process and essential condition for dilution	
	6.4. Self-purification of rivers and streams	
	6.5. Factors affecting self-purification Dilution, Current Sunlight Sedimentation, Temperature, Oxidation, Reduction,	
	6.6. Oxygen sag curve, introduction of Streeter Phelp's equation (Derivation not required), Numerical on self-purification of rivers/stream	
	6.7. Wastewater disposal by land treatment	

Specific Objectives	Contents	Duration in hours
	6.7.1. Suitability of land treatment,	
	6.7.2. Methods of land treatment -irrigation, overland flow and rapid infiltration, Broad irrigation and sewage farming, Methods of application of sewage on land flooding, surface irrigation, ridge and furrow method, subsurface irrigation and spray irrigation,	
	6.7.3. Sewage sickness and its prevention	
	Unit 7. Wastewater Treatment Method	12
- Knowledge about treatment objectives and different types. - Design consideration of different treatment process	7.1. Objectives of wastewater treatment	
	7.2. Treatment process types and impurity removal	
	7.3. Primary treatment process	
	7.3.1. Racks and Screens purpose and types	
	7.3.2. Skimming tank purpose and construction	
	7.3.3. Grit chamber ~ purpose, construction and design criteria	
	7.3.4. Sedimentation purpose, types and design criteria	
	7.3.5. Chemical precipitation purpose, mixing and flocculation (introduction only)	
	7.3.6. Numerical on design of Grit chamber and Sedimentation tank	
	7.4. Biological (Secondary) treatment process	
	7.4.1. Objectives of biological treatment process, Principles of biological treatment process, Attached and Suspended growth processes, Types of biological treatment process	
	7.5. Sewage filtration	
	7.5.1. Filter types	
	7.5.2. Intermittent sand filter purpose, construction, working and cleaning with merits and demerits	
	7.5.3. Contact bed purpose, construction, Working and cleaning with merits and demerits	
	7.5.4. Trickling filter purpose, construction, working and cleaning with merits and demerits, types, design criteria, Numerical on design of trickling filters	
	7.5.5. Activated sludge process	
	7.5.5.1. Principles of activated sludge process, Construction and process description, Aeration methods, Design criteria, Advantages and disadvantages, Sludge volume index, Numerical on activated sludge process	
	7.5.6. Oxidation ponds	
	7.5.6.1. Purpose of oxidation ponds, Theory of oxidation ponds, Construction of oxidation ponds, Commissioning, Operation and maintenance, Advantages and disadvantages, Design criteria, Numerical on oxidation ponds	
· Know about	Unit 8. Sludge Treatment and Disposal	4

Specific Objectives	Contents	Duration in hours
sludge and its ultimate disposal methods · Design consideration	8.1. Sources of sludge, Necessity of sludge treatment, Characteristics of sludge, Determination of sludge volume, Volume - Moisture relation	
	8.2. Sludge treatment methods	
	8.2.1. Grinding and blending, Thickening ,Gravity thickener, purpose, construction and loading criteria, Digestion Aerobic and anaerobic digestion, digestion process, control of digestion, construction and design criteria of digester, Dewatering Vacuum filtration (purpose and construction), Drying Sludge drying beds (purpose and construction) Composting ~ purpose, principles, types (windrow and mechanical) incineration purpose and construction	
	8.3. Numerical on sludge volume determination and design of digester	
	8.4. Sludge disposal methods	
	8.4.1. Dumping, Land filling, Lagooning, Spreading on land	
· Knowledge about, onsite sanitation, using it in emergency, disposal of sewage from isolated building Concept of rural sanitation, Eco sanitation, pit latrine · Design consideration	Unit 9. Disposal of Sewage from Isolated Buildings	4
	9.1. Necessity	
	9.2. Introduction of Rural sanitation	
	9.3. House Drainage , general principles , Pipes and Traps, classification of traps , introduction to system of plumbing	
	9.4. On site sanitation Definition and types, concept of using it in disaster (emergency)	
	9.5. Pit privy purpose and construction	
	9.6. Ventilated Improved Pit (VIP) latrine purpose, construction, design criteria, types (single pit, double pits and multiple pits), advantages and disadvantages	
	9.7. Pour flush latrine purpose, construction and design criteria	
	9.8. Septic tank purpose, construction, design criteria, working and maintenance	
	9.9. Septic tank effluent disposal methods	
	9.9.1. Drain field - purpose, construction and design criteria	
	9.9.2. Soak pit ~ purpose, construction and design criteria	
	9.9.3. Evapotranspiration mound ~ purpose and construction	
	9.9.4. Leaching cesspool purpose and construction	
	9.10.Numerical on design of VIP latrine, Pour flush latrine, Septic tank, Drain field and Soak pit	
	9.11.Introduction to Concept of Eco sanitation, concept of reuse of wastewater	
	9.12.Introduction to constructed wetland	
· Knowledge of Introduction	Unit 10. Solid Waste Disposal	3
	10.1.Introduction of solid waste ,Characteristic of solid waste	

Specific Objectives	Contents	Duration in hours
to solid waste and safe disposal · Know about different component and its ultimate disposal methods	10.2.Quantity of solid waste	
	10.3.Collection and transportation of solid waste	
	10.4.Solid waste disposal methods	
	10.4.1. Dumping, Sanitary landfill, Incineration, Composting	

References

1. B. C. Punmia and Ashok Jain, "Wastewater Engineering", Laxmi Publications (P) Ltd., and New Delhi.
2. P N. Modi, "Sewage Treatment & Disposal and Wastewater Engineering", standard Book House, Delhi.
3. G S Birdie and J S, Birdie, "Water Supply and Sanitary Engineering", Dhanpat Rai Publishing Company (P) Ltd" New Delhi.
4. KN. Duggal, "Elements of Environmental Engineering Ltd., New Delhi. S Chand and Company
5. Howard S. Peavy, Donald R. Rowe and George Tchobanoglous, "Environmental Engineering" McGRAW-Hill International Edition .

Far Western University
Bachelor of Engineering (Civil)
Course of Study 2071

Course Title: Airport and Railway Engineering
CourseCode.: CE 365
Year/Semester: Third/Sixth
Level: Bachelor of Engineering (Civil)

Credit: 3
Number of lecture/week: 3
Tutorial/week: 1
Total hours: 45

1. Course Introduction:

The main aim of this course is to provide a basic knowledge to the students to understand the fundamentals of Airport and Railway Engineering. At the end of this course, students will be able to design the basic elements of railway and airport engineering.

2. Course Objectives:

At the end of this course the student should be able to:

- understand the basic knowledge of railway and airport engineering.
- Be able to design the basic elements of railway and airport.

3. Specific Objectives and Contents:

Specific Objectives	Contents
• Understand transportation system components • Differentiate railway and airport engineering from highway engineering • Know the scope of railway and airport engineering	UNIT 1. Introduction to railway and airport engineering (4 hours) 1.1 Introduction to Transportation System 1.2 Scope of Railway Engineering 1.3 Scope of Airport Engineering
• Know the basic terminology of railway engineering • Recognize the components of railway and its requirements	UNIT 2. Railway Engineering (8hours) 2.1 Gauges 2.2 Problems with multigauge system 2.3 Permanent way 2.4 Wheels and Axles 2.5 Track resistance, Hauling capacity, Stresses in tracks and its components 2.6 Rails, creep in rails, failures in rails 2.7 Sleepers, Ballast, Fastenings

• Understand basic design elements of railway track • Design the basic components of geometrics of railway track • Know the state of the art technology used at present in railway engineering	UNIT 3. Geometric Design of Railway Track (12hours) 3.1 Railway track alignment 3.2 Horizontal curves and superelevation 3.3 Safe speed 3.4 Transition curve and extrawidening 3.5 Gradient and vertical curves 3.6 Turnouts and its design, crossings, junctions 3.7 Signals 3.8 Trains control system 3.9 Advancement in railway technology
• Understand the importance of air transportation • Know the concerned organizations and their scope	UNIT 4. Introduction to Airport Engineering (2hours) 4.1 Introduction to air transport 4.2 Organizations invoved in air transport (national and international) and their functions
• Know the basic elements of airport and aircrafts • Design the fundamental geometric elements of airport • Know the details of elements for airport planning and site selection • Understand the design concept of airport pavement	UNIT 5. Airport Engineering (19hours) 5.1 Airport classification 5.2 Aircraft characteristics 5.3 Aircraft controls, airport site selection, obstructions 5.4 Runway orientation, length, geometric 5.5 Taxiway 5.6 Aprons and aircraft parking 5.7 Terminal area and building 5.8 Hangers 5.9 Visual aids-markings, signals and signage 5.10 Airport pavement

Far Western University
Bachelor of Engineering (Civil)
Course of Study 2071

Course Title: Estimation, Costing and Valuation	Credit: 3
Course Code.: CE 366	Number of lecture/week: 3
Nature of the Course: Theory	Tutorial/week: 1
Year/Semester: Third/Sixth	Total hours: 60

1. Course Introduction:

The course is aimed to provide the basic knowledge of Estimation, Costing and Valuation of civil engineering works.

2. Course Objectives:

At the end of this course the student should be able:

- To understand the fundamentals of estimation, costing and valuation of civil engineering works.
- To estimate the quantities of item of works involved in buildings, water supply and sanitary works, road works and irrigation works,
- To do rate analysis of different items of work.
- To estimate the cost of various construction work.
- To do valuation of properties and preparation of reports for estimation of various items.

3. Specific Objectives and Contents:

Specific Objectives	Contents
• Understand the purpose of estimating, estimated cost and actual cost. • Understand the principle of units, measurement and payment of items of works.	Unit 1. Introduction (3 Hours) 1.1 Definition. 1.2 Purpose of estimating. 1.3 System of units. 1.4 Principle of units, measurement and payment for various items of works and materials.
• Know to estimate various types of estimate. •	Unit 3. Types of Estimate (5 Hours) 3.1 Approximate estimate 3.2 Detailed estimate 3.3 Revised estimate 3.4 Supplementary estimate 3.5 Annual repair and maintenance estimate 3.6 Complete estimate 3.7 Split up of cost of building and road works, water supply and sanitary works.

• Understand the purpose, importance, and requirement of analysis of rate. • Understand the factors affecting analysis of rate. • Know the factors affecting the analysis of rate. • Be able to prepare analysis of rate for building works, road works, irrigation works, water supply and sanitary works.	Unit 4. Analysis of Rates (10 Hours) 4.1 Introduction 4.2 Purpose analysis of rate 4.3 Importance analysis of rate 4.4 Requirement analysis of rate 4.5 Factors affecting the analysis of rate 4.6 Method of preparing analysis of rate for 4.6.1 Building works 4.6.2 Road works 4.6.3 Irrigation works 4.6.4 Water supply and Sanitary works
• Be able to estimate building, road, irrigation, water supply and sanitary works. • Be able to estimate joineries for panelled and glazed doors, windows, ventilators, handrails etc. • Be able to estimate Slab culvert, Retaining wall, Septic tank and Soak pit.	Unit 5. Detailed Estimate (20 Hours) 5.1 Estimate of two storied building: Calculation of quantities of Brick work, PCC, RCC, Plastering, Colouring, Painting, and Varnishing of flat and sloped roof. 5.2 Estimate of road works, irrigation works, water supply and sanitary works. 5.3 Estimate of joineries for panelled and glazed doors, windows, ventilators, handrails etc. 5.4 Estimate of Slab culvert, Retaining wall. 5.5 Estimate of Septic tank and Soak pit.
• Be able to understand and make valuation of property. • Be able to write a report of valuation of property.	Unit 6. Valuation(7 Hours) 6.1 Introduction. 6.2 Purpose of valuation. 6.3 Terms used in valuation: Capitalised value, Depreciation, Rent, Mortgage, and Lease. 6.4 Methods of determining value of property: Land and building. 6.5 Report writing method of valuation.
• Be able to estimate building, road, irrigation, water supply and sanitary works. • Be able to estimate Retaining wall, Slab culvert, Aqueduct, Septic tank and Soak pit.	Unit 7. Tutorial. (15 Hours) 7.1 Estimate of a two room building. 7.2 Estimate of a double storied residential building. 7.3 Estimate of a portion of hill road. 7.4 Estimate of a Retaining wall. 7.5 Estimate of a RCC slab culvert. 7.6 Estimate of an aqueduct. 7.7 Estimate of septic tank and soak pit.

Far Western University
Bachelor of Engineering (Civil)
Course of Study 2071

Course Title: Seminar
Course Code.: CE 367
3Year/Semester: Third/Sixth
Level: Bachelor of Engineering (Civil)

Credit: 1
Number of lecture/week:1
Total hours: 15

1. Course Introduction:

This course is aimed at teaching students about the preparation of seminar on the relevant topics of their interest in Civil Engineering.

2. Course Objectives:

At the end of this course the student should be able to:

- Conduct critical analysis on the relevant topics of their interest
- Develop presentation and communication skills in both oral and written form
- Enhance writing techniques and prepare report on the relevant topics of their interest.

3. Specific Objectives and Contents:

Specific Objectives	Contents
To highlight on significance of seminar, and different techniques used in conducting critical analysis and research activities and writing reports on the relevant topics of Students' interest in Civil Engineering	Introduction to Seminar; Significance of Research in Civil Engineering; Different Techniques and Skills of Presentation and Writing the Report on relevant topics (3 hrs)
To enhance writing and presentation skills of students, provide them appropriate feedbacks to improve their verbal skills, and make them able to face experts and large mass of audience.	Selection of relevant topics of student's interest; Preparation of reports; Presentation of relevant topics in front of subject experts and large mass of audiences; and Providing feedbacks to students about their presentation and report writing skills (12 Hrs)

Marks Distribution:

1) Student's attendance, class performance, participation in seminar discussion and learning attitudes=40%

2) Report Writing, Final Presentation, and Critical Analysis on the relevant topic of student's interest=60%

(Each student should present his/her research work/report verbally, and attend all the seminar presentation series by students)

Far Western University
Bachelor of Engineering (Civil)
Course of Study 2071

Course Title: Concrete Technology and Masonry Structures
Course Code: CE368
Year/Semester: Third / Sixth
Level: BCE (Bachelor of Civil Engineering)
Total Lectures: 45hrs

Credit: 3
Number of lecture/week: 3
Tutorial/week: 1
Lab/week: 2/2

1. Course Introduction

The main aim of this course is to provide theoretical as well as practical information on concrete technology and masonry structures. The first part of the course deals with concrete technology. In this part student will learn properties of ingredients of concrete and will be able to know concrete mix design. The students will also study properties of green and hardened concrete, factors affecting these properties and will learn the tools and techniques of quality control and quality assurance in different stages of their use. The second part deals with Masonry structures and in this part students will learn classification, construction technologies and behavior of masonry structures. The students will be able to analyze and design masonry structures for gravity loads and lateral loads. And at last they are able to perform test for concrete as well as masonry works.

2. Course Objectives

At the end of the course the students should be able to

- Different types of cements and its use, properties of concrete ingredients.
- Structure of concrete, concrete mix design and quality control.
- Properties of green and hardened concrete and its testing including nondestructive tests.
- Factors responsible for concrete durability.
- Classification of masonry, construction technology and behavior of masonry structure under gravity and lateral loads.
- Design of masonry wall for vertical load and lateral load.
- Testing of masonry units and walls.

3. Course Outline

Specific Objectives	Contents	Duration
Part A: Concrete Technology		
• Use of concrete and its type. • Properties of concrete ingredients • Types of cement and its use. • Classification of admixture and its use.	1. Introduction to concrete and concrete materials 1.1. Use of concrete in structure and types of concrete 1.2. Concrete materials - Role of different materials (Aggregates, Cement, Water and Admixtures) 1.2.1. Aggregates –Types and properties of aggregates and their gradation 1.2.2. Cement - Manufacturing of cement, Compound composition of Portland Cement, Structure and reactivity of compounds 1.2.3. Different types of cements and its use 1.2.4. Use of water in concrete 1.2.5. Admixtures - Classification of admixtures, Uses of admixtures in concrete	4hrs

• Structure of concrete • Concrete structure as three phase system	2. Structure of concrete 2.1. Concrete as three phase system 2.2. Structure of aggregate phase 2.3. Structure of the hydrated cement paste phase 2.4. Transition zone in concrete	3hrs
• Properties of concrete • Testing for workability • Quality control at site	3. Property of green concrete 3.1. Workability and its test 3.2. W/C ratio in concrete 3.3. Segregation and bleeding 3.4. Batching, Mixing, handling, placing, compaction and curing for quality concrete	3 hrs
• Nominal mix • Mix design	4. Mix design of concrete 4.1. Probabilistic concept in mix design approach 4.2. Introduction to nominal mix 4.3. Concrete mix design by DOE, ACI and IS Method	4 hrs
• Properties of hardened concrete • Shrinkage & creep • Fatigue strength • Porosity in concrete • Formation of gel	5. Properties of hardened concrete 5.1. Deformation of hardened concrete and Modulus of elasticity 5.2. Shrinkage and creep in concrete 5.3. Strength against fatigue, impact and dynamic loading 5.4. Effect water-cement ratio and aggregate size on porosity 5.5. Effect of gel/space ratio	4 hrs
• Strength and weakness of concrete • Test for different strength • Acceptance criteria • Non destructive test in concrete	6. Testing of concrete and quality control 6.1. Various strength of concrete: Tensile, Compressive, Shear and Bond 6.2. Compressive strength test 6.3. Tensile strength test 6.4. Variability of concrete strength and acceptance criteria 6.5. Non-destructing testing of concrete	4 hrs
• Factors affecting durability of concrete • Causes of concrete deterioration	7. Concrete durability 7.1. Effect of water and permeability on concrete durability 7.2. Physical and chemical causes of concrete deterioration 7.3. Carbonation and corrosion of steel in concrete 7.4. Concrete in extreme temperatures	3 hrs
Part B: Masonry Structures		
• Use of masonry as structure • Construction technology • Masonry units • Reinforcement in masonry	8. Introduction to masonry structures 8.1. Use of masonry structures 8.2. Construction technology - English bond, Flemish bond, Rat-trap bond 8.3. Hollow block and compressed earth block 8.4. Masonry as infill walls 8.5. Reinforced and un-reinforced masonry	4 hrs
• Design of masonry wall for gravity loads	9. Design of masonry walls for gravity loads 9.1. Introduction to codal provisions 9.2. Design example for gravity loads Solid wall, wall with openings, walls with eccentric loadings and walls acting as columns	6 hrs

• Performance of masonry wall under lateral loads • Design for lateral loads • Ductility in masonry structure	10. Masonry structures under lateral loads 10.1. Performance of masonry structures in lateral loads 10.2. Failure behavior of masonry structures in lateral loads 10.3. In-plane and out-of-plane behavior of masonry structures 10.4. Ductile behavior of reinforced and unreinforced masonry structures 10.5. Calculation of stresses for lateral loads 10.6. Elements of lateral load resisting system	6 hrs
• Testing of masonry units • Testing for compressive and shear strength of walls	11. Testing of masonry elements 11.1. Compressive strength of bricks and walls 11.2. Diagonal shear test 11.3. Non-destructive tests - Elastic wave tomography, Flat-jack, Push shear test and others	4 hrs

Experiments/Laboratory Works

Part I : Concrete Technology

1. Grading and Properties of aggregates
2. Concrete Mix design: Nominal mix, DoE, ACI and IS Method
3. Test of concrete cubes, cylinders, prisms
4. Non-destructive testing

Part II : Masonry Structures

5. Test of bricks on Compression
6. Test of wall on Compression

Demonstration of Non-destructive test

References:

1. A.M. Neville, J.J. Brook, Concrete Technology, International Students' Edition
2. M. S. Shetty, Concrete Technology: Theory and Practice, S. Chand, New Delhi, 2005
3. P.K. Mehta, Paulo j. M. Monteiro, Concrete, Microstructure, Properties and Materials, University of California, Berkley (Indian Edition)
4. A.S. Arya, Masonry and Timber Structures including earthquake resistant Design, Nem Chandra and Bros, Roorkee, 1987
5. A.W. handry, B.P. Sinha, S.R. Davies, An Introduction to Load Bearing Brick Design, University of Edinburgh, 1981
6. P. Dayaratnam, Brick and Reifnorced Brick Structures, Oxford and IBH Publishing Co. Pvt. Ltd. 1987
7. Ns 511:2060, NS 512 :2060, NS 513:2060, NS 516:2060 andNS 517:2060 OR Related NBC
8. IS 456:2000,IS 383:1970 (Reaffirmed 1997), IS 516:1959 (Reaffirmed 1999), IS 1905:1987 (Reaffirmed 2002), IS 4326:1993 (Reaffirmed 2003),IS 2212:1991 (Reaffirmed 2005)
9. SP 20:1991 and SP 23:1982

Semester VI	Course Code	22 Credits
Soil and Rock Mechanics	CE361	3
Irrigation Engineering	CE362	3
Design of Steel & Timber Structure	CE363	3
Sanitary and Environmental Engineering	CE364	3
Airport & Railway Engineering	CE365	3
Estimation, Costing & Valuation	CE366	3
Seminar	CE367	1
Concrete Technology & Masonry Structure	CE368	3